

Related Work Analysis for Context-Aware False Alarm Reduction in Violence Detection

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Abstract

This report reviews recent studies on video-based violence detection and identifies the research gap that motivates our proposed context-aware gating approach. Existing methods mainly focus on improving classification accuracy, computational efficiency, or interpretability. However, false alarm reduction is still not sufficiently addressed, even though false positives are a major issue in real surveillance systems.

1 Related Work

Recent studies on violence detection in surveillance videos have mainly focused on improving classification accuracy, computational efficiency, and model interpretability. CUE-Net [1] is one of the most relevant works because it addresses the same binary violence detection task on datasets such as RWF-2000 and RLVS. The method combines spatial cropping, an enhanced UniformerV2 backbone, and efficient additive attention to improve benchmark performance.

Several recent methods also investigate lightweight and gated architectures. Dual-Branch VideoMamba with Gated Class-Token Fusion [2] uses separate spatial and temporal branches and combines them through an internal gating mechanism. This shows that gating can be useful for violence detection. However, the gate is embedded directly inside the backbone, which means the whole model still needs to be trained as a complete architecture. Vi-SAFE [3] also focuses on efficient spatial-temporal modeling for public surveillance, showing that lightweight violence detection is important for real deployment.

Context-aware violence detection has also been explored. The CAET + ST-GCN framework [4] models human-environment interaction by using skeletal and temporal graph features. This approach is useful because it shows that context can support violence detection. However, it depends on pose estimation, detection, and tracking, which increases the computational cost and makes the system harder to deploy. Other explain-

able violence detection frameworks [5] highlight the importance of interpretability and generalization in surveillance scenarios.

Table 1: Comparison between recent violence detection methods and our proposed approach.

Method	Main Goal	Context Usage	Gating Mechanism	False Positive Focus
CUE-Net [1]	Accuracy improvement	Not explicitly used	No	No
VideoMamba-GCTF [2]	Accuracy and efficiency	Not explicitly used	Internal backbone gate	No
Vi-SAFE [3]	Efficient surveillance detection	Not explicitly used	No	No
CAET + ST-GCN [4]	Context-aware localization	Yes, but costly	No	No
Explainable CNN [5]	Interpretability and generalization	Not explicitly used	No	No
Ours	False-alarm reduction	Annotation-free scene context	External lightweight gate	Yes

Although these studies improve violence detection performance, several limitations remain. First, most methods mainly optimize overall accuracy rather than explicitly reducing false positives. This is a practical issue because false alarms can overload human operators and reduce trust in surveillance systems. Second, scene context is often ignored or extracted using expensive pipelines such as pose estimation and tracking. Third, many approaches require replacing or retraining the full backbone, which limits their flexibility when applied to existing violence detectors.

To address these limitations, our project proposes a two-stage, model-agnostic framework for false-alarm reduction. Instead of replacing the base detector, we use a frozen X3D-S violence classifier and add a lightweight context gating module. This module uses annotation-free scene cues, including crowd density, lighting condition, and motion coherence, to estimate a gate weight α that adjusts the final violence prediction. This design aims to reduce false alarms while keeping the system lightweight, interpretable, and reusable across different datasets.

2 Research Questions

Based on the identified gap, this project focuses on three research questions. First, we investigate whether X3D-S with context gating can reduce the false positive rate compared with X3D-S alone. Second, we examine which context stream contributes most to false-alarm reduction. Third, we evaluate whether the proposed gating framework can generalize from RWF-2000 to RLVS.

3 Conclusion

The reviewed studies show that recent violence detection research has made progress in accuracy, efficiency, context modeling, and interpretability. However, explicit false-alarm reduction remains insufficiently addressed. Our proposed lightweight context gating module is designed to fill this gap by improving deployment reliability without retraining the entire backbone.

References

- [1] Senadeera et al., “CUE-Net: Violence Detection Video Analytics with Spatial Cropping, Enhanced UniformerV2 and Modified Efficient Additive Attention,” in *CVPR Workshops*, 2024.
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